AI at Jefferies

Current Activities, Perspectives and Elements of a Strategy

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AI – Introduction to Concepts

Artificial Intelligence (AI) is a vast discipline that has been used across many industries and applications for many years (the term was coined in 1956). Within AI, **Machine Learning (ML)** involves automating tasks and simulating behaviors typically performed or exhibited by humans or complex systems.

In the **Financial Services (FS)** industry, AI/ML have been used predominantly in Risk Management (such as fraud detection), and Wealth Management (such as robo-advisory), and conservatively to support Trading and Investment Banking.

There are two approaches to ML:

- With **Supervised Learning**, systems use large sets and extensive combinations of known inputs/outputs as data points. These combinations constitute what is referred as the "Ground Truth". The systems learn (or are "trained") from these data. This approach is by-and-large statistic-driven and is typically used for classification or forecasting purposes (e.g., buying behavior by customer segment).
- With **Unsupervised Learning**, systems find patterns or generate output from a set of inputs alone, i.e., without the explicit prior knowledge of what constitutes a 'correct answer' or "Ground Truth". This is typically used in decision support systems, chess playing engines and NLP/LLMs defined below.

In the last decade, both approaches have benefited from the introduction of **Neural Network** architecture that mimics how brains function and to do so, uses thousands of parameters incrementally modified as more training data is leveraged, a process known as "learning".

Natural Language Processing/Understanding (NLP/NPU) use both approaches to understand and generate human language. NLP techniques are used in a wide range of applications, from virtual assistants and chatbots to sentiment analysis.

Artificial Intelligence (AI), has been used for many years, but Chat GPT has opened it up to many more people, and it is only the tip of the iceberg.

ChatGPT is based on a type of ML called **"Large Language Models" (LLMs).** LLMs are new in that they use hundreds of billions of parameters and are trained in exceptionally gigantic data sets (a significant subset of the Internet). In addition, they use refinements of how knowledge is represented and processed to generate text (Generative Pre-trained Transformer).

ChatGPT is the tip of the iceberg: Other similar LLM offerings (including by Google and Facebook) are based on the same approach, while an ever-expanding ecosystem of vendors and open-source developers has created tools to build private versions of GPT-based LLMs, processing various public and/or proprietary sources.

See Supporting Material A – An Introduction to Artificial Intelligence for more details.

Current State of AI at Jefferies

At Jefferies, AI/ML is used to extract, synthesize, and summarize data faster, to support better decisioning, to optimize task execution through automation and to enhance collaboration. Our approach has been to lead with business problems, developing solutions pragmatically and in a risk-conscious way.

On the **Madison Program** we have used AI/ML to develop a range of solutions for Equities and Investment Banking to source more demand, uncover net new client interests, and generate more trades for the Firm.
Pure ML applications include:

- **EQ CIL (Client Interest List)** – Provides scores to the US/EU Equities Cash Desk that describe how a client is likely to be interested in trading a specific symbol, based on a range of factors specific to the client's profile and past trading patterns.
    - The desk is provided two ratings, one based on a descriptive model (the "Reality Model"), the other one based on cutting edge machine learning techniques (the "Predictive Model").
    - The two outputs are correlated, however; the two models inherently differ in that the Reality Model is used to find which clients are historically interested in a given stock while the Predictive Model is used to predict "who is going to trade today".
    - Both models use a combination of trading activity, market data, risk factors, interactions, holdings information, as well as readership data and ultimate decisions to recommend trades are left to the desk's judgement.
- **IOI Routing Model** – Offers the US Central Risk Book (CRB) and Electronic Trading Services (ETS) teams well-targeted recommendations on which clients should be invited to respond to both buyer and seller interest. Recommendations are based on previous behavior patterns without posting the IOI publicly, as this would leak information on the buyer overall strategy to all market participants.
- **Activism** - Predicts the probability that a company will be the target of an activist campaign (used by the Activist Defense Team).

Broader AI techniques routed in more traditional statistical and rule-based methodologies on the **Madison Program** include:

- **II Campaign** - Enables Equity Research to identify contacts to target to vote for our Analysts based on historical behavior/consumption of research from/interaction with Jefferies.
- **Contact Interest Model** - Provides Research Analysts and Sales with recommendations to drive Content Research Distribution and Corporate Access Allocation (used by Research Sales and Analysts in US and EU)
- **Madison Link** - Leverages bankers digital communications with clients, such as meetings and emails (extracting only relevant, available information such as banker name and client name, but not the full content of the communications), to uncover relationships with and across potential and existing clients and provide users with a 360-degree view of relationship data between Jefferies and clients (used by IB, FI & EQ)
- **IB Deal Screens** - Assess prospective clients' likelihood to consume Jefferies services in the leveraged buy outs and convertible issuances domains (used by ECM Structured Products (converts)/M&A Teams (LBO) Teams).

In **Fixed Income**, the AI models built by the business to support market making include:

- US Algo Trading
    - **Demand model** - An alpha-generating predictive signal for short-term buy/sell imbalances in credit markets, critical for overcoming selection bias in making markets
    - **Latent factor model** - A model to identify hidden factors driving the market, and each bond's sensitivity to these factors
    - **Flow imbalance model** - A model to estimate future volume and buy/sell imbalance by bond, useful for estimating future liquidity opportunities and potential future risk imbalances
    - **Market predict** - A model to estimate real-time bid and offer for corporate bonds, continuously calibrated to match the market
    - **RFQ edge** - A model that takes into account the current state of the market, nature of the request for quote, and the demand state of the bond to estimate the edge (expected profit) if we were to trade the request
- European Rates
    - **Predicting client flows** - Past client flows are used to build a pattern of client behavior and predict client flows. Patterns can be around auctions or around events like month end. Initial results have been encouraging. For some clients where we see regular flows, we have been able to pinpoint exactly which bonds they are going to sell on a given day with over 75% accuracy
    - **European Rates Auction prediction model** - Traditional auction strategy models have broken down as Hedge Funds have exploited these strategies and now try to front run the auction process. Hence, we need a ML/adaptive process to benefit from auction strategies. We use tick level data around auctions to predict the relative cheapening/richening of bonds around auctions and relative value trade opportunities
- **European Credit Algo Trading** uses various non-linear ML models to predict several key components of our IG systematic market making platform, for instance:
    - Where the next bond trade is likely to happen as a correction to available composite prices
    - Our probability of trading given the characteristics of RFQs sent by clients
    - The relative future performance of bonds to control the idiosyncratic risk in the book and generate alpha

In **Corporate**, we use Behavox, an AI-driven compliance solution for electronic communications surveillance. to detect evidence of collusion, harassment, or insider trading, and sends summarized reports to the Compliance Team for review/action.

Emerging AI Activities in Financial Services and in Fintech

In the FS industry, the use of AI continues to expand across Risk Management and Wealth Management with new efforts emerging in Trading, Investment Banking and Research, some of which we are also exploring:

- In **Risk Management**, AI/ML is being used to help financial institutions assess risk more accurately by analyzing large amounts of data and identifying potential risks before they become a problem. In Compliance, it can help financial institutions to comply with regulations by analyzing data and identifying potential compliance issues. This can help companies to avoid fines and other penalties for non-compliance
- In **Wealth Management**, AI will provide more personalized, data-driven advice in specific areas such as portfolio optimization, risk management, fraud detection, tax analysis and relationship management. On the retail side, Wealth Management for the mass affluents will expand using generative AI LLMs in in digital assistant forms. AI-enhanced new entrants and AI-driven new robo-advisors' solutions might significantly challenge established market participants.
- In low-touch **Trading**, frequency trading is a common use case but as of now, this uses rule-based logic, a somewhat primitive level of AI.

- The growth of AI-powered transaction platforms in **Investment Banking**, will enable bankers to analyze and assess client behavior/activity and identify trends or insights on future banking needs. Bankers can leverage AI on transactional due diligence documentation and risk analysis.
- In **Research** some of the techniques used to harvest the underlying knowledge have been known for a long time, but the scale of data and data processing now involved in the cloud has delivered stunning results. AI will reshape traditional research, because it can synthesize a phenomenal amount of information into immediately usable artifacts that can be used easily
- Generally, every investment organization has a unit that leverages (mostly supervised) Machine Learning, has made investments in **Data Science** models that aim at identifying reliable patterns to guide investment decisions or algo trading. Some companies (e.g., Chase) have organized **large R&D group** to support ambitious goals commensurate with their size (large) and their product mix.

In both FS and **FinTech**, ChatGPT and related announcements (by Facebook, Google, and AWS) have received massive attention and ChatGPT has been hyped as a game changer. This said, as our analyst (See AI Deep Dive - Thill Jefferies (Research - May 2023.pdf) notes:

- *"AI is not new, but the consumer facing version of ChatGPT has opened people's eyes.*
- *Enterprise adoption of AI remains early as businesses are in the experimental phase with betas and limited previews, presenting investors with unclear business models and monetization plans.*
- *Overall, we see the best positioned groups to include hyperscalers, other enablers & data clouds, and large, differentiated platforms.*
- *We see headwinds such as trust, accuracy, IP protection, and bias as some of the key hurdles for the industry*

Both established market participants and startups are incorporating LLMs in their product offering. Most notably, Bloomberg started strong with BloombergGPT, and we should expect Bloomberg's or a competitor's LLM to become the standard digital assistant of Wall Street traders. The BloombergGPT case is important as AWS claims that it was entirely based on SageMaker, their cloud ML platform, and while Microsoft has won the PR battle with a first-mover advantage, AWS has also announced new LMM-specific offerings which are gaining traction.

See Supporting Material B – Industry Current State and Ongoing Developments for more details.

Emerging AI Activities at Jefferies

At Jefferies, as well as expanding our existing efforts, we are considering additional use cases being explored by others in the FS industry (previous section), but in a way that is consistent with our size and overall strategy.

The (evolving) domain where AI/ML is being considered at Jefferies includes:

- In **Global Research** we are exploring Microsoft Azure OpenAI (the enterprise OpenAI ChatGPT product) for:
  - Summarization of transcripts e.g., earnings calls.
  - Sentiment analysis of e.g., conferences and earnings calls (as input to Research and IB). Sentiment analysis is subjective, but the technology can scan a transcript and infer that the "tone" is positive, negative or neutral based on the text from the transcript and what the model has been trained on
  - Translation capabilities (including transcripts referenced above).
- In **Investment Banking** we are looking at solutions to foster collaboration and streamline how banking teams operate:
  - Using LLMs for internal search capabilities on available, relevant data on entities such as companies and products in Call Reports to seek synergies across accounts.

    - Generating image suggestions for Pitch Books
- In **Fixed Income** we are looking into proof of concepts to leverage the latest generative AI capabilities:
    - (US & European Algo Trading) Automate quant coding leveraging GitHub Co-Pilot
    - (US Algo Trading) Scrape product mentions and client inquiries from real time Bloomberg IB Chat data; Create a chatbot to generate answers to pricing and inventory related questions to free up traders time
    - (Global Credit) Scape product mentions and client inquiries from real time Bloomberg IB Chat data via BloombergLLM model
    - (European Credit) LLMs to transform textual data related to companies, securities or client interactions into new structured data that can be consumed by our ML models
- For **Developers and Quants** GPT should be able to automate 20-40% coding in the next few years.
- In **Corporate**, Jefferies Data Service (JDS) is an asset and a differentiator, which we are leveraging to explore AI solutions.
    - We are looking to use NLP and LLM to:
        - Automate Operations - AI could help in generating automated emails to clients and counterparties, with guardrails in the form of human review.
        - Improve User Experience, through interfaces with HR, Compliance and Tech Support e.g., Chatbot or automatic assistance in using forms.
    - We are exploring general ML-supported Analytics in:
        - Operations in the context of T+1 roll-out, to infer and mitigate risk of execution delays by counterparties based on previous performance.
        - Compliance to review communications to analyze and predict employee behaviors
        - Finance to review anomalies and breaks on our Balance Sheet and infer potential future anomalies that could be symptomatic of financial loss potential.
    - We are actively engaged with Behavox to test their Quantum AI tool which will give more advanced results with less errors and higher quality alerts for our electronic communications surveillance.

Across the firm, **Software Development** will be impacted by LLMs that have now demonstrated they can generate or assist in the generation of code and automation scripts in a reliable and efficient way. We are working on implementing the proper controls given the legal, compliance and security risks at play (See section on Risk and Compliance Approach below).

One of the possible solutions (being investigated now in terms of suitability and cost) is a version of ChatGPT which would run in an unshared environment. This service offering from Microsoft is Azure OpenAI. We are defining a new governance framework for the **Microsoft Azure OpenAI enterprise ChatGPT** product to enable it at Jefferies, with an initial focus on specific use cases (mostly for content summarization and translation), which are under review.

Overall, key AI/ML initiatives are being performed by existing market participants. More importantly innovation, particularly innovation delivered by **startups** is built on the **Cloud**, leverages **Big Data** and is built using **AI/ML** as part of its DNA.

At Jefferies, we will continue to identify opportunities and use AI to develop our own solutions, in a safe environment.

The **future of FS** will be significantly bound by these developments. At Jefferies, we will continue to identify opportunities and not only participate in the new industry dynamics but shape them through the development of our own solutions, in a safe environment.

Risk and Compliance Approach for AI at Jefferies

AI solutions are data-intensive, highly connected systems for which risk and security issues (such as data security, privacy, bias) are well known and not specific to AI or ML. However, these risks will be exacerbated by AI becoming more pervasive within the enterprise.

In addition, AI introduces a new series of risks:

- AI outcomes can be inaccurate and not suitable for consumption by clients e.g., if communications commit the firm to AI-generated research or recommendations.
- AI might generate output that is based on information that is copy- or trademark-protected.
- AI might generate highly realistic yet fraudulent output.
- AI might generate unpredictable yet realistic results that cannot be relied upon, for instance, when generating text, systems can associate concepts and references in a way that creates unfounded conclusions, so called "hallucinations"

With little or no regulatory framework in place for now, the right compliance approach for AI at Jefferies is to continue with our existing human-in-the-loop processes. These processes provide guardrails against the unfiltered propagation of AI-generated information and ensure our people are accountable for using judgement in all decision making.

This said, AI-powered digital assistants will continue to be added within the enterprise to the knowledge worker's overall supporting working ecosystem. As with any information sources, bankers, traders, analysts etc will continue to be ultimately accountable for using judgement in decision making – this is the general approach Jefferies has been using for all AI/ML initiatives in production today.

See Supporting Material C – Compliance and Risk/Threat Management for more details.

Supporting Materials

A – An Introduction to Artificial Intelligence

Artificial Intelligence (AI) is a broad collection of techniques that has been used across many industries and applications for many years, from voice recognition and spam filtering to product recommendations, chatbots and fraud detection. **Machine Learning (ML)** is a specific area within AI that aims at automating tasks and behaviors typically performed by humans or complex systems.

ML systems detect patterns in data and generate text, sounds, images, or actions that mimic human-like behavior. ML systems continuously modify how they respond to stimuli by learning from the data they are exposed to and encoding this evolving knowledge in ML "models".

There are two categories of ML, Supervised Learning and Unsupervised Learning.

- With **Supervised Learning**, Models use an extensive combination of known inputs/outputs as data points. These combinations constitute what is referred as the "Ground Truth". The systems learn (or are "trained") from these examples to generate correct outputs for any other input. It is typically used for classification or forecasting purposes (i.e., buying behavior by customer segment) in a way that is deeply rooted in statistical sciences.
- Models generated through **Unsupervised Learning** are designed to find patterns or generate output without the explicit prior knowledge of what constitutes a 'correct answer', that is without Ground Truth

ML (in both categories) has been around for decades, but its ability to perform increasingly sophisticated tasks is due to dramatic changes in computing and storage performance and cost, and the stupendous amount of data available to train Models.

Another noteworthy progress has been fueled by significant advances in algorithmic sciences, specifically, the use of so-called **Neural Networks** (for both Supervised and Unsupervised Learning). Neural Networks simulate how brains function: outputs are derived by feeding inputs into a "layered" mesh of simple calculations units ("neurons") that uses weighted sums to generate outputs. The weights (or Model Parameters) typically number in the thousands (or hundreds of thousands at the most) and are constantly adjusted based on outcome performance – in fact, this feedback loop is what constitutes "learning".

In the last few decades, achievements using Neural Network-based Unsupervised Machine Learning have been truly stunning: Google's AlphaZero routinely outperforming all Grand Masters and language translation such as Google Translate that are approaching perfection.

The most recent (< 3 years) announcements concern a type of ML Model called **"Large Language Models" (LLMs)**. LLMs are new in that they use billions of parameters and are trained in truly gigantic data sets (a significant subset of the Internet data). In addition, they use refinements of how knowledge is represented and a new type of Model architecture **(Generative pre-Trained Transformer)** that produces not just one input at a time but a series of outputs. This how **ChatGPT** generates textual responses to human requests in the form of whole sentences and paragraphs.

Similar solutions (based on the same architectures) have been introduced by others since ChatGPT and it is important to note that none of these can be unreservedly relied upon – fundamentally, they can only regurgitate information they were trained on and the process can introduce errors, including so called "hallucinations" where inaccurate or irrelevant data use corrupts the veracity of the answers provided. In addition, ChatGPT has been trained with data only available up to 2021, but this is likely to be improved in the future.

B – Industry Current State and Ongoing Developments

Risk Management

- Fraud detection: AI algorithms are being used to analyze copious amounts of data and identify patterns that indicate fraudulent activities
- Risk management: AI is helping financial institutions assess risk more accurately by analyzing copious amounts of data and identifying potential risks before they become a problem
- Compliance: AI can help financial institutions to comply with regulations by analyzing data and identifying potential compliance issues. This can help companies to avoid fines and other penalties for non-compliance
- Multifactor authentication (voice / face recognition)

Wealth Management

- For the mass-affluent, Robo-Advisors still use predicate logic to implement "modern" portfolio theory (introduced in the 50s) and pricing models (like Black Sholes, introduced in the 60s)
- Predicate logic is also used for systems delivered internally by banks aiming at advising the Mass Affluents
- Chatbots will continue to grow in customer service to enable advisors to focus on more high-value, high-touch services to their clients
- AI-powered chatbots and virtual assistants will provide customers with personalized assistance in real-time. Beyond the financial advisory space in Wealth Management this is not that mature in financial services
- The democratization of wealth management will expand through the use of generative AI LLMs
- AI will provide more personalized, data-driven advice in specific areas such as portfolio optimization, risk management, fraud detection, tax analysis and even relationship management
- Investment and Asset Management are actively looking for ways to leverage ChatGPT today to enable advisors to:
- Quickly access customer data and answer customer queries
- Provide customers with more personalized advice and targeted investment advice

Trading

- AI has been used to help drive strategic trading decisions, at the discretion of the trader
- In low-touch trading, frequency trading is a common use case. It is rule-based and uses primitive levels of AI and does not involve traders
- Algo trading routinely uses various models that aim at identifying reliable patterns to guide investment decisions
- AI-powered natural language dialogue could help front, back, and middle office trade functions. Traders and operations analysts can ask questions about regulatory reporting on historical trades to identify trading inefficiencies, fix regulatory gaps, provide real time querying capabilities, and ensure timely settlement
- Only a few financial firms have tried and succeeded in using reinforcement learning (RL), an approach that is at the centered on how ChatGPT learns to enable machines to surpass humans e.g., the IB Deep Blue chess engine. This could be effective in quantitative finance and money management
- In Financial Markets Infrastructure AI-powered chatbots can serve exchanges, trade depositories, and central clearinghouses. Client-side users can receive live trade support and expedited onboarding. Internal analysts can utilize AI trained on internal data to drive positional risk management

Investment Banking

- With an AI-powered transaction platform, investment banks can analyze and assess client behavior/activity and identify trends or insights on future banking needs. Bankers can leverage AI on transactional due diligence documentation and risk analysis.

Research

- Some of the techniques used to harvest the underlying knowledge have been known for a long time (n-gram analysis, Recurrent Neural Network), but the scale of data and data processing now involved in the cloud has delivered stunning results
- The technology will reshape traditional research (the same way it will reshape the legal profession), because it can synthesize a phenomenal amount of information into immediately usable artifacts that can be used easily
- This said, here like in other areas, we can expect that there will continue to be a need for human contribution to the research process of course to guide the likes of ChatGPT in the right direction – analysist will become more editors than creator though.

Other Market Participants / FinTech

- ChatGPT is the tip of the iceberg: Other similar LLM offerings (including by Google and Facebook) are based on the same approach, while an ever-expanding ecosystem of vendors and open-source developers has created tools to build private versions of GPT-based LLMs, processing various public and/or proprietary sources.
- For a comprehensive analysis of the market dynamics, see recent Jefferies research: AI Deep Dive - Thill Jefferies (Research - May 2023.pdf

C – Compliance and Risk/Threat Management

The nature and extent of the societal risks emerging in recent months due to new AI developments are likely to shape the risk management and compliance frameworks that companies will be compelled to adopt, either through regulation-driven or voluntary adoption, particularly (bot exclusively) for activities affecting the public at large (through social networks for instance) and in Europe.

Although fundamental, these risks are not discussed here. We focus on risks that are or are likely to impact companies with no core retail business.

In the business realm we need to consider AI solutions as data-intensive, highly connected systems for which risk and security issues (such as data security, privacy, bias introduced by using unrepresentative data sources) are well known and not specific to AI or ML. These risks will be exacerbated by AI becoming more pervasive within the enterprise. For example, data generated by an internal AI model can be exfiltrated to unauthorized parties, so data leakage prevention techniques should be expanded. That said, it is reasonable to expect that the policies and techniques used to manage these risks and adhere to compliance/regulatory frameworks are likely to be applicable to the AI domain, with AI-specific features but through existing conceptual structures (such as Definition or Risk, Policies, Remediation).

What AI is introducing, however, is a genuinely new series of risks related to the intrinsic nature of AI in general and ML in particular:

- AI can create outputs that can be difficult to verify or explain. Sharing the unfiltered outcomes of a model can create a host of issues if the communication commits the firm, in delivering AI-generated research or investment recommendations
- AI might generate output that is based on information that is copy- or trademark-protected with the firm being unaware of such risk.
- AI might generate highly realistic yet fraudulent outputs that can trigger a new class of risks. Posting an outrageous message on social media using a fake address/handle is one thing, making that a deep-fake video realistically impersonating a company CEO can be substantially worse.

- AI can generate unpredictable yet realistic results that cannot be relied on – For instance, when generating text, systems can associate concepts and references in a way that creates unfunded conclusions, so called "hallucinations".

For now, it is unclear if the magnitude and frequency of these risks will be an insurmountable impediment to the deployment of client/public-facing AI systems. Previous generations of AI systems (like face recognition) were quite fallible at first before outperforming humans in for instance, detecting individual faces in a sea of people.

FURTHER NOTES

1) **Jefferies Approach to CoPilot and Azure OpenAI** - Check out the Enterprise Architecture (EA) site forum page - See: https://jefferies.service-now.com/community?id=community_home and answers on the topic there: then click on the top question: *"What are we with ChatGPT - What can we expect in general in terms of developer support and also general use of LLM?"*

2) **Research from Jefferies** - J-Research on CNBC from Brent Hill, our Technology Analyst.
   - Highlights (See AI Deep Dive - Thill Jefferies (Research - May 2023.pdf):
     - *"AI is not new, but the consumer facing version of ChatGPT has opened people's eyes.*
     - *Enterprise adoption of AI remains early as businesses are in the experimental phase with betas and limited previews, presenting investors with unclear business models and monetization plans.*
     - *Overall, we see the best positioned groups to include hyperscalers, other enablers & data clouds, and large, differentiated platforms.*
     - *We see headwinds such as trust, accuracy, IP protection, and bias as some of the key hurdles for the industry and expect regulation will follow*
     - *Least at Risk from AI: MSFT, GOOGL, AMZN, META, SNOW, ADBE, INTU, PWSC, PCOR, CRM, PANW, HUBS*
     - *Tweener Positioning in AI: ORCL, TEAM, GDDY, WIX, ESTC, FRSH, DOCU, UDMY, DDOG, PLTR, WDAY, NOW*
     - *Most at Risk from AI: CHGG, DBX, SEMR, MSTR, LAW, FIVN, NICE"*
   - Also, see:
     - https://www.youtube.com/watch?v=xQVL8gU-v3M
     - https://www.youtube.com/watch?v=5e1ey8dbme8

3) **Who is Doing What and Where in FS** -- Every investment organization that has a unit that leverages statistical machine learning, has made investments in Data Science, especially in buy-side where talent is not an issue:
   - **Morgan Stanley** has a group for machine learning research lead by Yuri Nevmyvaka who is a former student of Michel Kearns. Together they co-authored one of the earliest papers on reinforcement learning for optimal trade execution (stocks not bonds)
   - Manuela Veloso is currently leading AI Research at **JP Morgan** and has been publishing on reinforcement learning
   - **Goldman Sachs** has a core AI team that aims at leveraging deep learning to solve quantitative finance problems. Options market makers within investment banks started relying heavily on the "approximation" power (not the predictive angle) of deep neural network to speed up real time pricing of large portfolio of options contracts.
   - **Citadel** have Li Deng, a Fellow of the Academy of Engineering of Canada as their Chief AI Officer and Global Head of Machine Learning
   - **DE Shaw** hired Pedro Domingos, Head of Machine Learning Research
   - **Bridgewater** have David Ferrucci, Director of Applied AI, but limited by the environment there
   - On the FinTech side, and with the arrival of LLM, **Bloomberg** started strong with BloombergGPT and we should expect Bloomberg's or a competitor's LLM to become the standard digital assistant of Wall Street traders

4) Other References

- **Books** - A pivotal moment in promoting Machine Learning (ML) as a very important subset of AI occurred when Igor Halperin and Matt Dickson, published their book on ML/RL applied to quantitative finance and Hans Buehler (JP Morgan) won 2022 quant of the year for his work on deep hedging, which uses deep RL for OTC derivatives hedging. Both works generalize existing work with RL, which means RL (and therefore ML) is no longer an alternative approach but is now proven to be a generalization that remedies shortcomings and hurdles facing the standard approach.
- https://www.weforum.org/agenda/2023/04/future-jobs-2023-fastest-growing-decline/
- https://www.businesswire.com/news/home/20230426005383/en/Alphabet-CEO-Highlights-Behavox-Google-Partnership-Recognizes-PaLM-AI-Adoption-in-Q1-2023-Earnings-Call
- https://www.cnbc.com/amp/2023/03/14/morgan-stanley-testing-openai-powered-chatbot-for-its-financial-advisors.html
- https://reports.jpmorganchase.com/investor-relations/2022/ar-ceo-letters.htm
- https://www.bloomberg.com/news/articles/2023-04-21/vanguard-fidelity-experts-explain-how-ai-is-changing-wealth-management
- Bloomberg Article_"Wall Street Banks Are Using AI to Rewire the World of Finance" May 31, 2023. Highlights below:

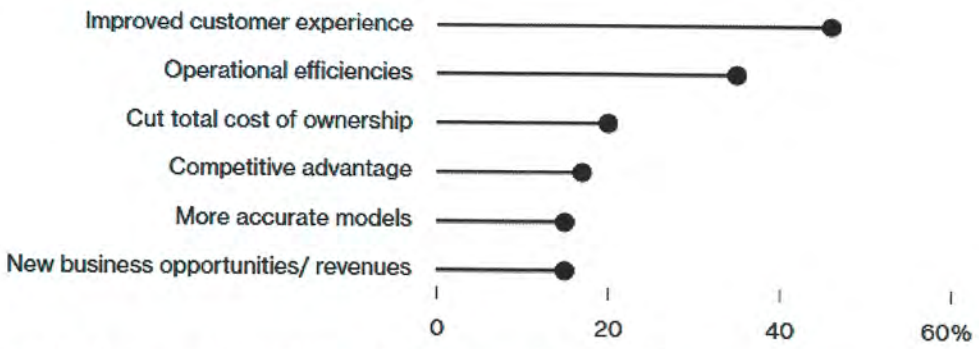
What You Need to Know About AI Technologies

Type of AI	What it does	Example
Neural networks	Learn through trial and error, similar to a human brain	Facial recognition technology, automated translation systems
Machine learning	Feeding data into algorithms so they get more refined over time	Used in finance for fraud detection, credit scoring and portfolio management
Deep learning	Layers of neural networks taught to classify large datasets	High-frequency trading, algo trading, market forecasting
Natural language processing	Helps understand, process and generate speech and text	Enables banks to analyze news sentiment, regulatory filings, social media data
Predictive analysis	Uses historical and real-time data to forecast future outcomes	Helps Wall Street banks identify trading opportunities and optimize investment strategies
Generative AI	Creates visual or text works from simple prompts	Image and text generators Stable Diffusion, Midjourney, Dall-E
Large language models	Very large networks trained on massive amounts of text	The best-known LLMs are GPT-3 or GPT-4, which drive ChatGPT

Source: Bloomberg reporting

Financial Services Firms Say AI Benefits Customer Services

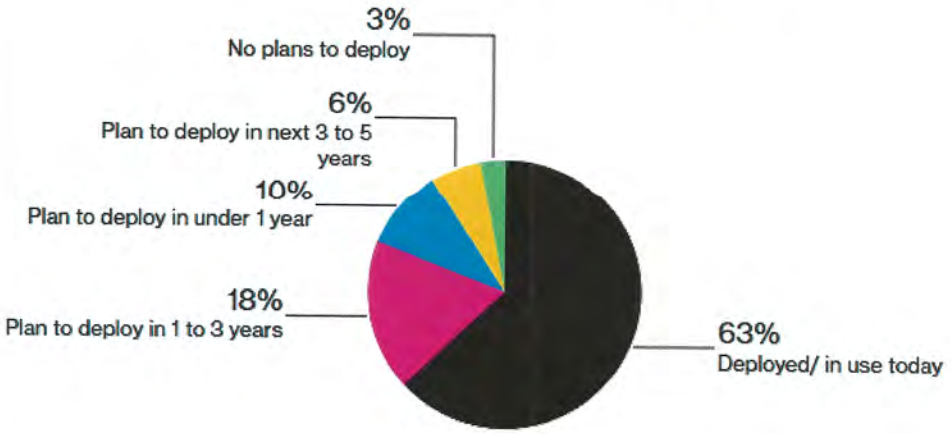
Almost half of firms cite AI as a way to improve experiences



Source: Nvidia Corp.'s 2023 State of AI in Financial Services Report
Nvidia surveyed approximately 500 global financial services professionals in firms including banks, asset managers and insurers.

Most Banks and Investment Firms Are Already Using AI

Percentage of respondents at each stage of introducing robotics



Source: 2022 Gartner Data and Analytics Transformation Survey
Gartner surveyed 101 firms. The data were published in December and do not include those who responded 'not sure' or 'unaware.'